Lab 2 – BevOS Requirements and Specifications

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1. Introduction

1.1 Purpose

This Software Requirements Specification specifies the requirements for the BevOS prototype. Its purpose is to describe, in enough detail for the system to be designed, implemented, and tested, what BevOS must do and the qualities it must meet. It records the functional, performance and non-functional requirements agreed upon by Team Silver and serves as the authoritative reference for building and evaluating the prototype. The intended audience includes BevOS development team and the instructor of the course (Professor Kennedy).

1.2 Scope

BevOS is a centralized, web-based workflow management platform designed for beverage manufacturing and co-packing operations. The beverage industry currently coordinates production through spreadsheets, email chains, phone calls, and disconnected software, which creates communication delays, approval bottlenecks, and reduced operational visibility.

BevOS addresses this by bringing brands, distributors, suppliers, co-packers, logistics providers, and internal operational teams into a single platform. Its goals are to improve operational efficiency by reducing repeated manual coordination, reduce communication and approval delays, increase visibility across the production lifecycle, organize operational documents within their related workflows, and support data informed management decisions. The system provides a modular foundation that scales with business growth rather than replacing every existing system.

1.3 Definitions, Acronyms, and Abbreviations

Approval Workflow — A process used to review and approve operational tasks before they move to the next stage.

BevOS — A centralized workflow-management platform designed for beverage manufacturing operations.

Client — An organization or individual purchasing services or products from another participant in the manufacturing process.

Co-Packer — A third-party company that manufactures and packages beverages for another company.

Core Services — The shared BevOS system software layer that processes requests, applies system rules, coordinates functional modules, and manages data exchange.

Dashboard — A visual interface displaying operational information and performance metrics.

Distributor — An organization responsible for delivering products from manufacturers to retailers or customers.

ERP (Enterprise Resource Planning) — Software that integrates core business processes into a unified system.

Logistics — The planning and coordination of transportation and delivery activities.

MFCD (Major Functional Components Diagram) — A diagram illustrating the primary hardware and software components of the system.

Nice Package Company — The beverage-industry client and domain partner for whom BevOS is being developed.

PostgreSQL — An open-source relational database management system used to store application data.

Role-Based Access Control — A security method that grants system permissions according to the user's assigned role and responsibilities.

Single Source of Truth — A trusted, centralized set of current information used consistently by authorized stakeholders.

Stakeholder — A person or organization that participates in or is affected by a beverage-manufacturing workflow.

Workflow — A sequence of business activities completed to achieve a specific objective.

1.4 References

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1.5 Overview

The remainder of the SRS is organized into two major sections. Section 2, Overall Description, presents the product perspective, its primary functions, the characteristics of its users, and the constraints, assumptions, and dependencies that affect development. Section 3, Specific Requirements, presents the detailed functional, performance, and non-functional requirements for the BevOS prototype.

2. Overall Description

2.1 Product Perspective

BevOS is a new, self-contained web application rather than a component of a larger existing system. It is organized using a Major Functional Components Diagram (MFCD) structure that separates user interaction, shared application logic, operational modules, and persistent data storage. Users access the system from internet connected desktop or laptop running a modern web browser. The BevOS Web UI sends requests to BevOS Core Services, which applies workflow and access rules and coordinates the operational modules (Quotes, Communications, Production, Logistics, Document Storage, and Reporting). These modules read and write structured data through a PostgreSQL database. Because BevOS coordinates information workflows rather than operating production equipment, the prototype requires no specialized manufacturing hardware.

2.2 Product Functions

At a high level, BevOS provides the functions below.

Workflow Management: Coordinating foundation that ties requests, approvals, documents, communication, production, and logistics into one process.

Product Request Management: Creating, reviewing, and tracking new product requests.

Quote Management: Tracking supplier quote requests and associating them with product requests.

Approval Tracking: Recording approval status, responsibility, and history for workflow decisions.

Production Management: Coordinating schedules, milestones, and progress updates.

Logistics Coordination: Tracking shipment and delivery status tied to production.

Document Management: Storing operational files within their related workflows under access control.

Communication Platform: Keeping messages tied to the request, quote, or activity they concern.

Reporting and Dashboards: Summarizing operational data into role-appropriate views.

Issue Tracking: Recording workflow problems, owners, and resolution.

Role-Based Access Control: Limiting what each user can see and do by role.

2.3 User Characteristics

BevOS serves several distinct user groups with differing needs and technical expectations. All groups are assumed to interact with the system through a web browser and to hold the permissions appropriate to their role.

- **Brands** initiate product requests and monitor their progress; expected to be comfortable with standard web applications but not technical specialists.
- **Distributors** coordinate product availability and movement.
- **Suppliers** receive requests, provide quote information, and support material or service needs.
- **Authorized operational personnel** (production, logistics, and finance teams) contribute the information needed to complete each workflow.

2.4 Constraints

- BevOS is a web-based application accessed through a modern web browser; no native desktop or mobile client is provided.
- The system is developed in Python using the Django web framework and stores data in a PostgreSQL database.
- The prototype is deployed through Docker and Docker Compose, orchestrating the application's backend, frontend, migration, and database services.
- As a one-semester course prototype, BevOS implements a demonstrable subset of the full product; non-target modules may be represented by stubs rather than complete implementations.
- The prototype is intended for demonstration in a teleconference setting rather than production deployment.

2.5 Assumptions and Dependencies

- Users are assumed to have reliable internet access and a device capable of running a modern web browser.
- The prototype assumes its scope is limited to workflows within the beverage manufacturing and co-packing industry.

- Test data is assumed to stand in for live logistics and production activity during demonstration.
- The system depends on the availability of its server, application, and PostgreSQL database environment.
- The system depends on the Django framework and its associated Python libraries, deployed via Docker.